AP 158: FINAL PROJECT

Feedbacks of Western North Pacific Tropical Cyclone Properties on ENSO

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Background

TROPICAL CYCLONES IN THE WESTERN NORTH PACIFIC

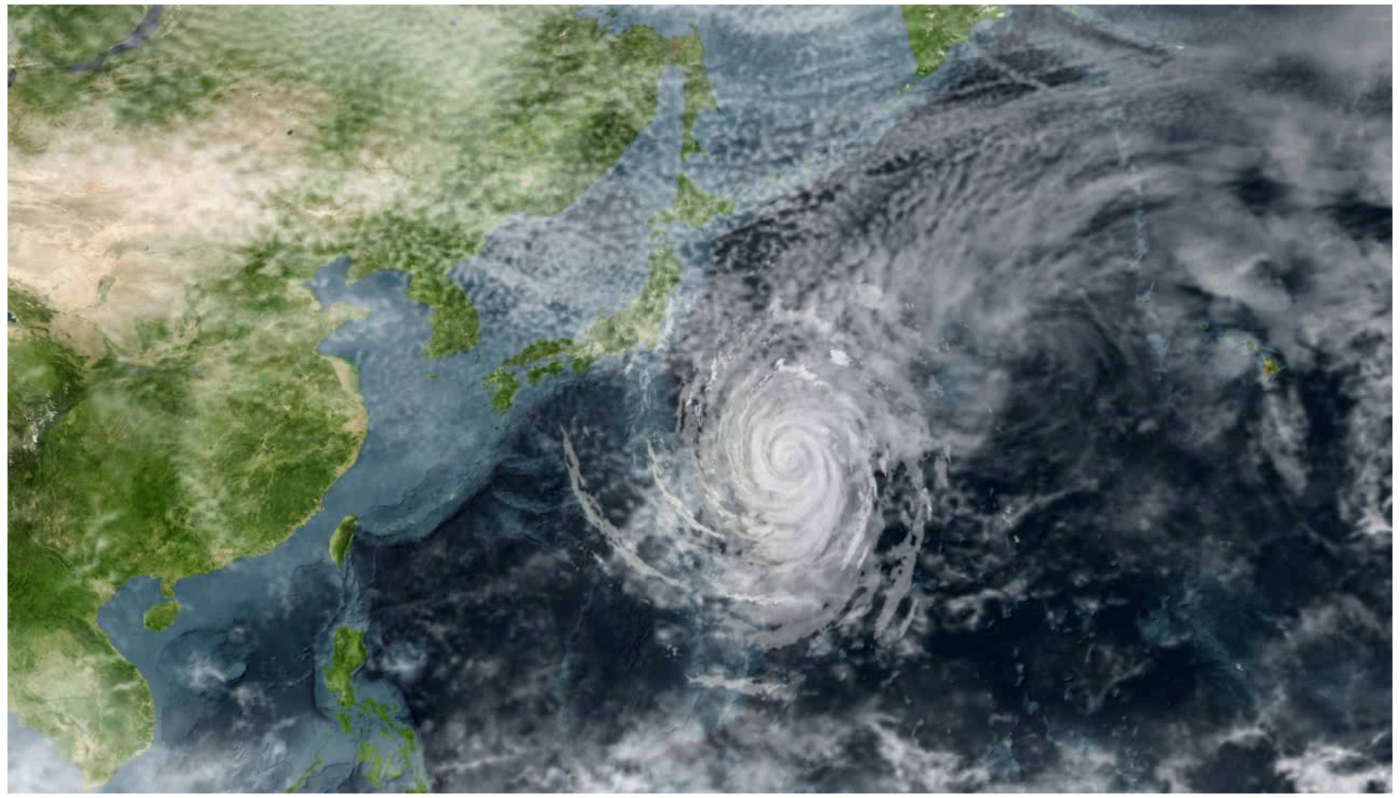

Figure 1. A mature TC that develops near Asia.
Note. From *Getty Images*, by Marvin Pineda, n.d.

Figure 2. Lightning during a thunderstorm at a village.
Note. From *Pexels*, by Michael Bird, n.d.

Tropical cyclones (TC) are warm-core low-pressure wind systems that bring torrential rains and extreme flooding, resulting in heavy casualties to human life and destruction to crops and properties in TC-prone countries like the Philippines.

Background

TROPICAL CYCLONES IN THE WESTERN NORTH PACIFIC

Among the seven basins over which TCs originate, the Western North Pacific (WNP) basin is the **most active**, accounting for nearly one-third of the total global TC occurrences. (Zhan et al., 2012).

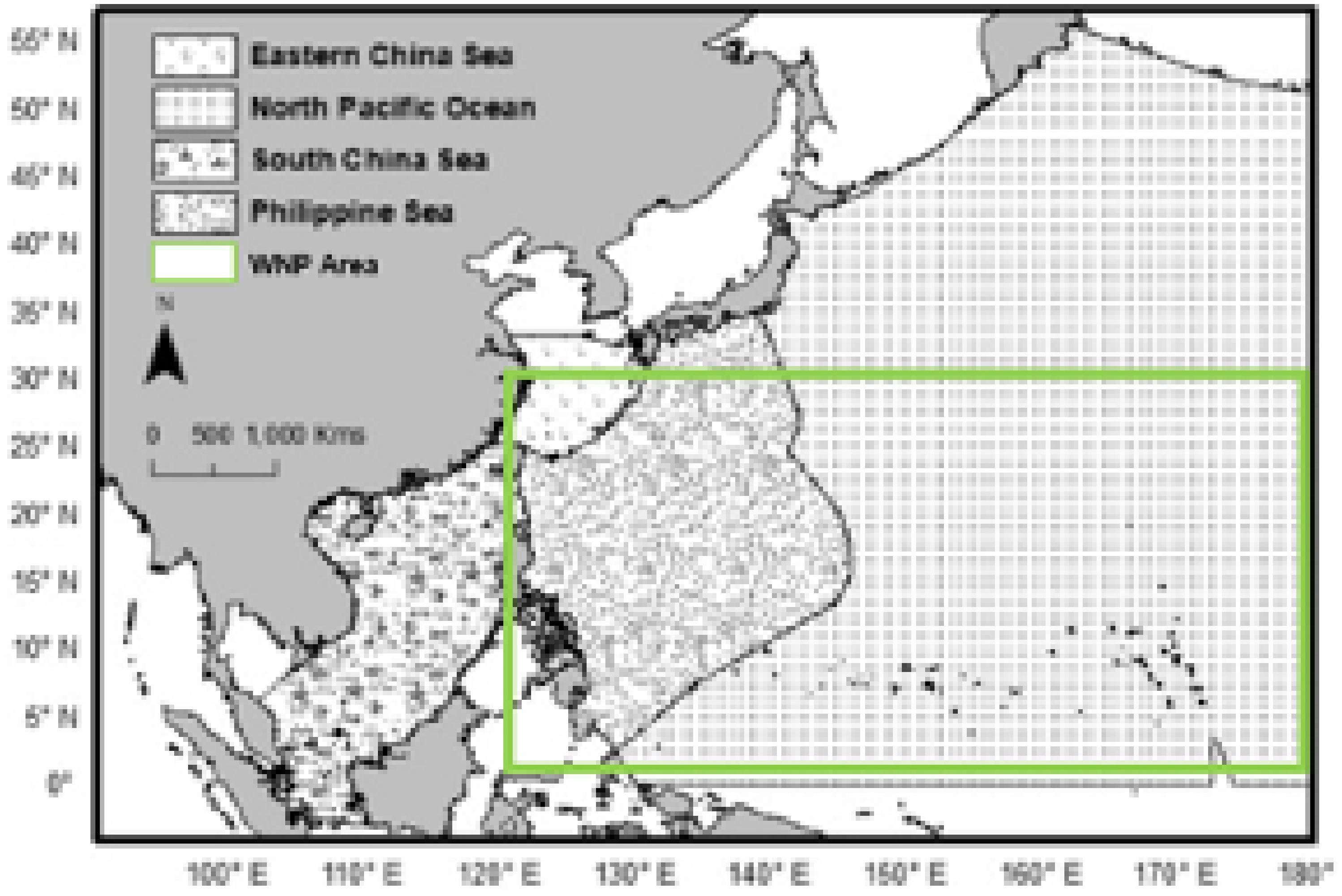

Figure 3. Ocean boundaries around the Western North Pacific Ocean. Adapted from Pandey and Liou (2022).

Background

El Niño-Southern Oscillation (ENSO)

- ENSO is the **periodic warming and cooling pattern in sea surface temperatures (SSTs)** in the central and eastern tropical Pacific Ocean, having a neutral and two extreme phases, namely La Niña and El Niño.
- Due to ENSO being the strongest interannual signal on the earth's climate system, many authors have studied its **influence on interannual variability of TCs over the WNP**. It has also been utilized as one of the main predictors in statistically based seasonal forecasts. (Chan et al. 1998; 2001).

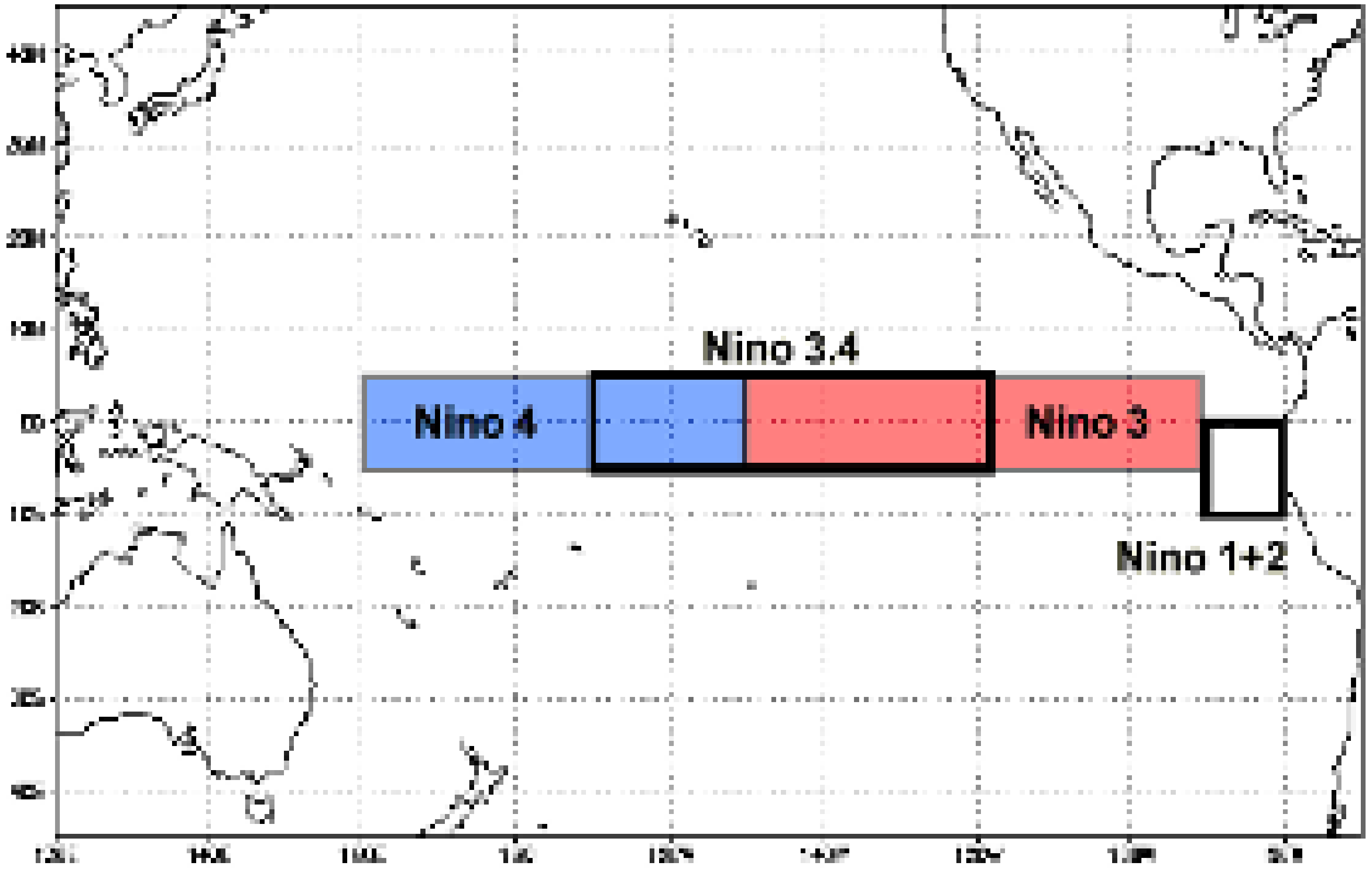

Figure 4. The 4 Niño regions (i.e., Niño 1+2, 3, 3.4, and 4) used to track SSTs in the central and eastern tropical Pacific Ocean. Reprinted from NOAA National Centers for Environmental Information (NCEI).

Objectives

Given the geographic location where ENSO persists, the study aims to provide insights on how such a climate phenomenon affects the properties of tropical cyclones that formed in the Western North Pacific basin over time. **Specifically, it aims to assess the feedback of TC frequency, TC days, TC intensity, and quarterly ACE on each ENSO phase.**

Hypotheses

Null: There is no significant relationship between each ENSO phase and the selected WNP TC properties (i.e., TC frequency, TC days, TC intensity, and quarterly ACE).
Alternative: There is a significant relationship between each ENSO phase and the selected WNP TC properties (i.e., TC frequency, TC days, TC intensity, and quarterly ACE).

Data Collection and Sorting

ENSO Phase

The Oceanic Niño Index (ONI) values from NOAA National Weather Service (NWS) were presented into 12 three-month periods:

DJF: December-January-February
JFM: January-February-March
FMA: February-March-April

MAM: March-April-May
AMJ: April-May-June
MJJ: May-June-July

JJA: June-July-August
JAS: July-August-September
ASO: August-September-October

SON: September-October-November
OND: October-November-December
NDJ: November-December-January

The data were sorted into four quarters: **JFM**, **AMJ**, **JAS**, and **OND**, by taking the average of the ONI values of DJF to FMA, MAM to MJJ, JJA to ASO, and SON to NDJ, respectively.

Data Collection and Sorting

ENSO Phase

Each quarter was classified as La Niña, neutral, or El Niño phase using the following intervals:

La Niña	**ONI < -0.5**
Neutral Phase	**$-0.5 \leq$ ONI < 0.5**
El Niño	**ONI $\geq$ 0.5**

(Corporal-Lodangco et al., 2016).

Data Collection and Sorting

Tropical Cyclone Data

- The TC data used were collected from the **Western North Pacific** (WNP) **Ocean best track data** of the **Joint Typhoon Warning Center** (JTWC).
- For the study, only the TC data from **1981 to 2021**.
- The following data were extracted from the JTWC data sets:
    - **maximum windspeed**
    - **maximum windspeed datetime**
    - **cyclogenesis** (cyclone formation) **datetime**
    - **cyclolysis** (cyclone deterioration) **datetime**
- Only TCs with **windspeed ⩾ 35 knots** (or 18 m/s) were considered (excludes weak Tropical Depressions (TD), which can cause zero ACE results).
- A total of 1065 TC data were extracted and sorted per quarter and per phase.

Standardization

The TC metrics were standardized using the following formula:

$$z = \frac{(x - \mu)}{\sigma}$$

z = standardized value

x = obtained TC data

μ = long-term mean (1981-2021 average) of the TC data

σ = standard deviation of the TC data

The standardization rescaled the TC data such that the mean is zero and the standard deviation is one.

Statistical Analysis

- To test for normality, the **normalitytest()** MATLAB function was used.
- To test for statistical significance between the results of TC metrics and the various ENSO phases, the **two-sample t-test** at the **95% confidence level** was performed.
  - The **ttest2()** MATLAB function was used.
  - In this test, the data is assumed to have normal distributions with unknown and unequal variances.

TC Metrics Analysis

TC Frequency
The average number of tropical cyclones per quarter and phase.

TC Days
The difference between the extracted cyclogenesis and cyclolysis datetimes.

TC Intensity
The average maximum windspeeds per quarter and phase.

TC ACE
The sum of the squares of maximum TC windspeeds summed over the quarter.

$$ACE = 10^{-4} \sum V_{max}^2$$

Standardization
The mean of the quarterly standardized TC metrics.

Bar Graph
Used to present the distribution of the mean of the quarterly standardized TC metrics.

Normality Test
Using the normality() function in MATLAB.

Two-Sample t-test
Employed for hypothesis testing, using the ttest2() MATLAB function.

TC Frequency

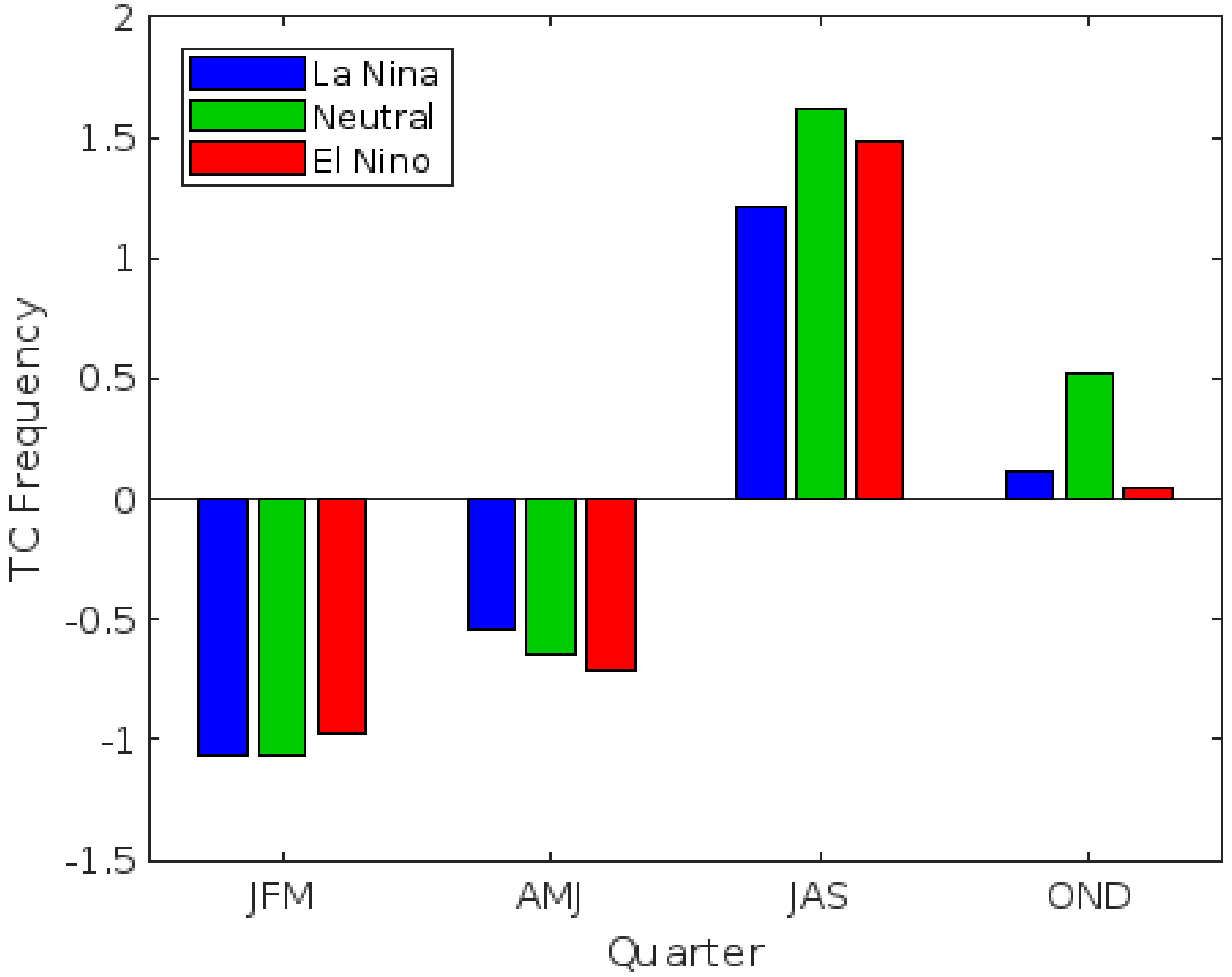

Figure 5. Mean scores of standardized quarterly TC frequency over WNP in different ENSO phases

Table 1. Summary of observation per quarter and season of TC activity.

Quarter	Season of TC Activity	Observations
JFM	early	Least TC count
AMJ	mid	Increase in TC count relative to the early season
JAS	peak	Most TC count
OND	late	Decrease in TC count relative to the peak season

Similar outcomes were observed by Zhao et al. (2011), Zhan et al. (2012), and Corporal-Lodangco et al. (2016) on the frequency of TC occurrence during the different TC activity seasons and quarters.

TC Frequency

Table 2. Result of the two-sample t-test for the TC counts per quarter and ENSO phase pairs.

		JFM	AMJ	JAS	OND
1	La Nina & Neutral	0	0	0	0
2	La Nina & El Nino	0	0	0	0
3	Neutral & El Nino	0	0	0	1

- There is a significant difference at the 95% confidence level in TC frequencies between neutral and El Niño years for the OND quarter.
- All other pairs of ENSO phase for all the other quarters obtained no significant difference.
- ENSO has no overall impact on the TC frequencies within the WNP domain.
- This result coincides with the observations of Lander (1994), Landsea (2000), Camargo and Sobel (2005), Zhan et al. (2011), Wu et al. (2019), and Zhao and Wang (2019), that ENSO causes a **negligible effect** on the **total TC frequency in the WNP basin** due to its **opposing effects on the eastern and western quadrants of WNP**.
    - During El Niño events, eastern WNP quadrants experience more TC occurrence, whereas western WNP quadrants experience the opposite (i.e., less TC occurrence) (Lander, 1994; Wang and Chan, 2002). Consequently, during La Niña events, the opposite occurs.

TC Days

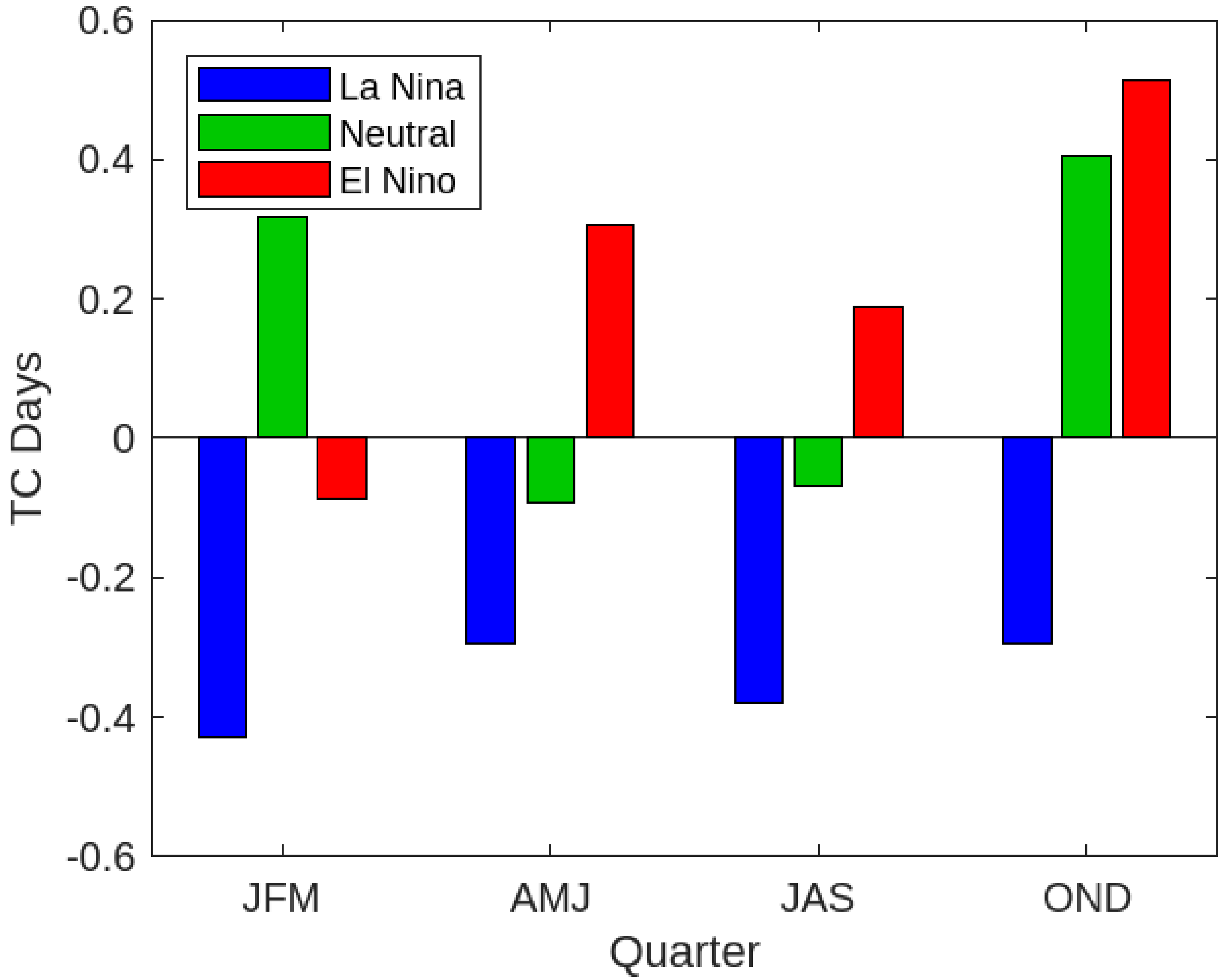

Figure 6. Mean scores of standardized quarterly TC days over the WNP basin during the ENSO phases: La Niña (blue), neutral (green), and El Niño (red).

- In El Niño phases, TC days are **above average** in all quarters except in JFM because the mean TC genesis location is **farther east and have the widest longitudinal range** among the three phases, allowing TCs to live longer while tracking westward of WNP.
- In La Niña phases, TC days are **below average** in all quarters. TCs during La Niña years **form closer to land**, so shorter periods of time are spent over the ocean before making landfall.
- In neutral phases, TC days are above average in JFM and OND, and marginally below average in AMJ and JAS, which can be attributed to the wide longitudinal range of TC genesis locations.

TC Intensity

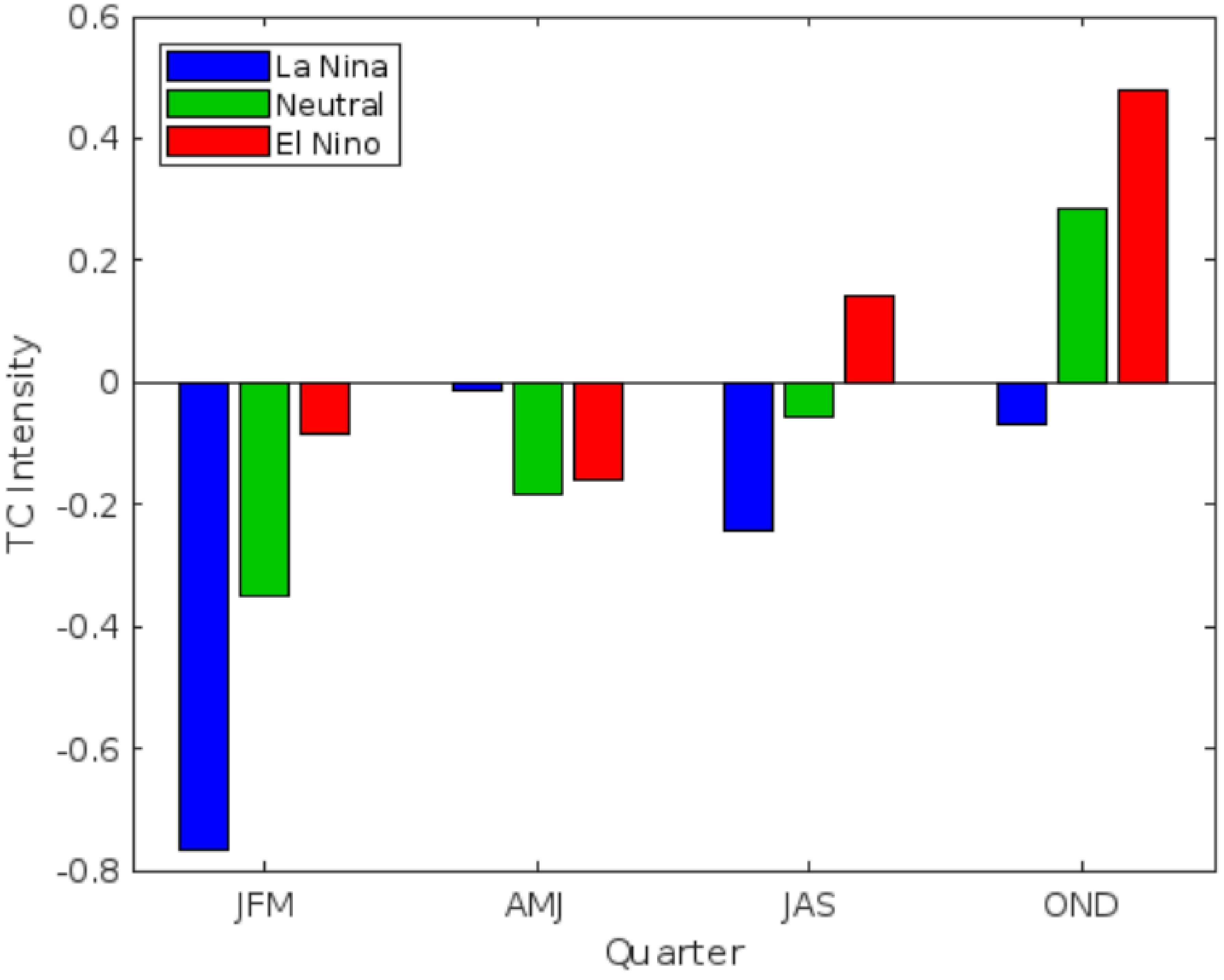

Figure 7. Mean scores of standardized quarterly TC intensity over WNP in different ENSO phases

- Intensity is **below average** in all quarters for **La Niña phases**. This is due to their short-lived nature which hinders their capacity to have stronger sustained winds.
- Significant difference only exists on the last two quarters. Here, **intensities are higher during El Niño phases** because of a southeastward shift in cyclogenesis location.
- In neutral phases, TC intensities are below average on the first two quarters and above average on the last two quarters.

TC ACE

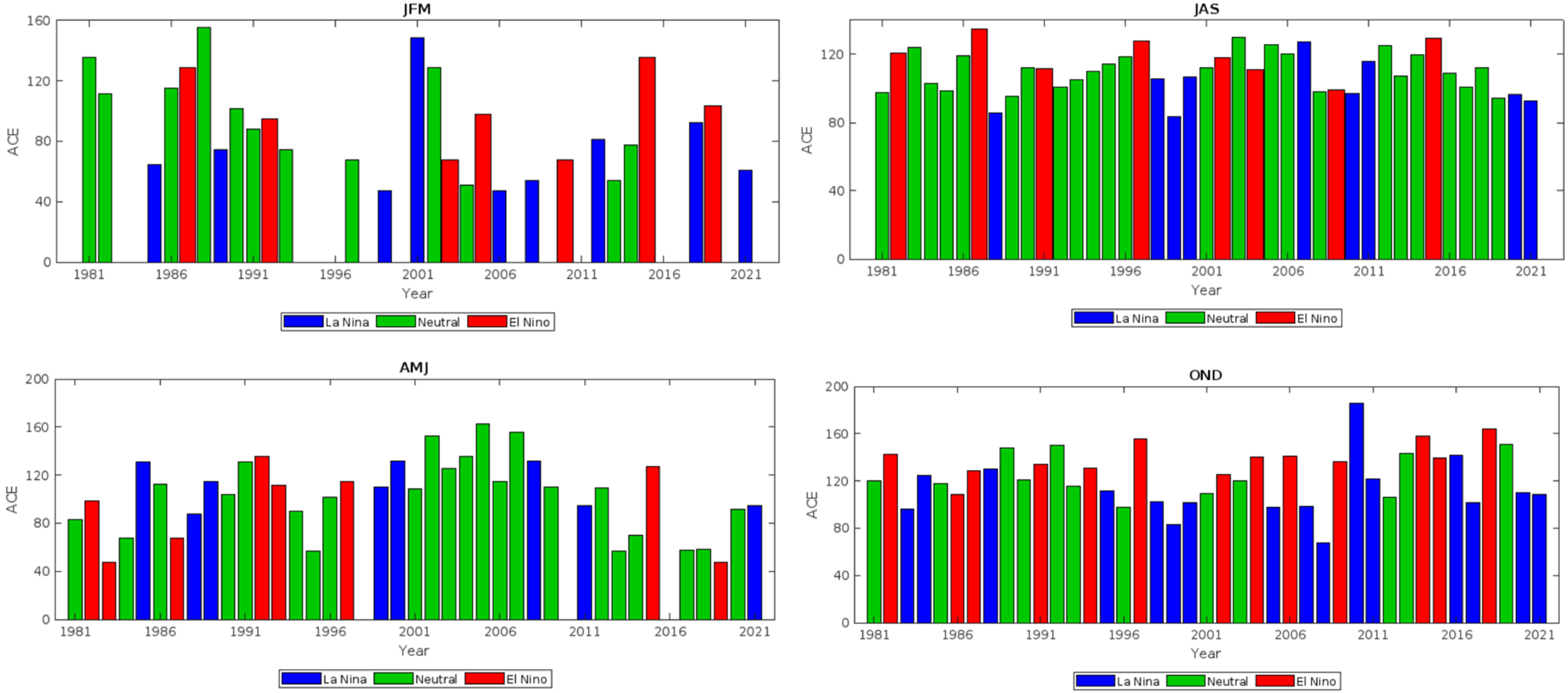

Figure 8. ACE annual time series for quarters JFM, AMJ, JAS, and OND

TC ACE

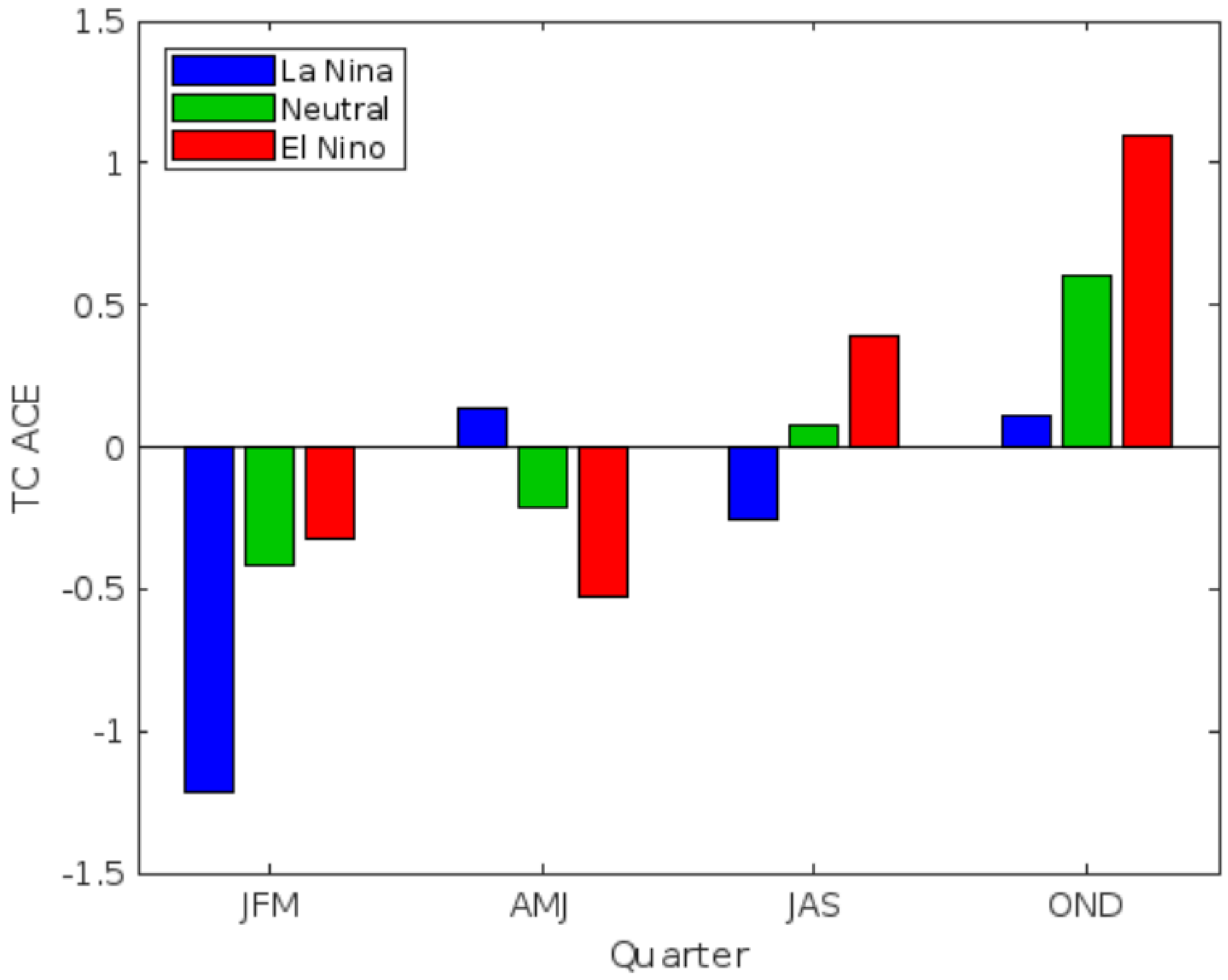

Figure 9. Mean scores of standardized quarterly TC ACE over WNP in different ENSO phases

- ENSO's effects on TC days and intensity manifested on TC ACE. The accumulated cyclone energy is **higher during the El Niño phases of JAS and OND** where TC days and intensity are shown to be high. This behavior is also influenced by the southeastward shift in cyclogenesis location.
- The above average intensity and days of TCs during the neutral phases of JAS and OND also affected ACE's value.
- ACE has the **lowest values during La Niña phase**, except for AMJ.

Conclusion

- ENSO has negligible impact on TC frequency due to the opposing effects of ENSO on the eastern and western quadrants of WNP.
- Longer TC lifetimes were observed in El Niño years due to a southeastward shift in cyclogenesis location. Shorter TC days were seen in La Niña phases.
- ENSO does not have a year-long effect on TC intensity. Statistical differences were only noted during the JAS and OND quarters. In these periods, intensities are weaker in La Niña phases and stronger in El Niño phases.
- ACE's feedback is affected by ENSO-influenced properties (TC days and TC intensity). As such, similar trends were observed. Quarters with higher (lower) TC days and intensity also garnered higher (lower) ACE.

Recommendation

- The regions of interest (Niño 3.4 region and WNP basin) are in opposite ends of the international dateline. Given this, use lag correlation analysis to check if there exists a time delay in the transmission of ENSO effects.
- Statistical schemes used in old studies (1970's-1980's) were reliant on the data available back then. With newfound results, the relationship between TC properties and ENSO indices can be used to update regression equations.
- Explore other TC properties like cyclogenesis location, TC tracks, and landfall probability.
- Analyze TC intensity based on TC classification (e.g., tropical storm, super typhoon).
- Forecast the feedbacks of TC metrics based on obtained TC-ENSO relationships.

Thank you very much!

References

Camargo, S. J., & Sobel, A. H. (2005). Western North Pacific tropical cyclone intensity and ENSO. Journal of Climate, 18(15), 2996-3006. https://doi.org/10.1175/jcli3457.1

Corporal-Lodangco, I. L., Leslie, L. M., & Lamb, P. J. (2016). Impacts of ENSO on Philippine Tropical Cyclone Activity. Journal of Climate, 29(5), 1877-1897. doi:10.1175/jcli-d-14-00723.1

Lander, M. A. (1994). An Exploratory Analysis of the Relationship between Tropical Storm Formation in the Western North Pacific and ENSO. Monthly Weather Review, 122(4), 636-651. doi:10.1175/1520-0493(1994)122<0636:aeaotr>2.0.co;

Landsea, C. W. (2000). El Nino/Southern Oscillation and the Seasonal Predictability of Tropical Cyclones. In H. Diaz, & V. Markgraf, EL NINO AND THE SOUTHERN OSCILLATION Multiscale Variability and Global and Regional Impacts.

National Weather Service (NWS). (n.d.). Climate Prediction Center - ONI. Retrieved from NOAA National Weather Service: https://origin.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php

References

Wu, R., Yang, Y., & Cao, X. (2019). Respective and combined impacts of regional SST anomalies on tropical cyclogenesis in different sectors of the western North Pacific. Journal of Geophysical Research: Atmospheres. doi:10.1029/2019jd030736

Zhan, R., Wang, Y., & Tao, L. (2014). Intensified impact of East Indian Ocean SST anomaly on tropical cyclone genesis frequency over the western North Pacific. Journal of Climate, 27, 8724-8739. doi: 600 10.1175/JCLI-D-14-00119.1.

Zhan, R., Wang, Y., & Ying, M. (2012). Seasonal Forecasts of Tropical Cyclone Activity Over the Western North Pacific: A Review. Tropical Cyclone Research and Review, 307-324.

Zhao, H., Wu, L., & Zhou, W. (2011). Interannual Changes of Tropical Cyclone Intensity in the Western North Pacific. Journal of the Meteorological Society of Japan, 89(3), 243-253. doi:10.2151/jmsj.2011-305

TC Metrics Analysis

TC Frequency

- The number of tropical cyclones per quarter and ENSO phase was tallied and averaged.
- The tallies were then standardized and graphed.
- The two-sampled t-test was also employed.

TC Days

- The difference between the extracted cyclogenesis and cyclolysis datetimes were considered their lifetimes.
- Similar to TC frequency, the results were averaged, standardized, and graphed on a per quarter and per phase basis.
- Then the two-sample t-test was employed.

TC Metrics Analysis

TC Intensity

- The extracted maximum windspeeds were averaged and standardized for each quarter and ENSO phase.
- The results were graphed and the two-sample t-test was employed.

TC Quarterly Accumulated Cyclone Energy (ACE)

- To represent TC intensities in a quarterly time scale, the quarterly ACE index values for each year were calculated using the following formula:
- The annual time series of quarterly ACE values were also graphed for each quarter (JFM, AMJ, JAS, OND).